

PASS

EN 1993-1-1 (UK NA) — fully restrained beam member check 457 191 82 UB S275

Governing: **Bending $M_{Ed}/M_{c,Rd} = 0.912$** (ULS: 1.35G + 1.5Q (Eq 6.10))

Shear 0.3 • Bending 0.91 • Defl 0.61

Member Loading and Member Forces

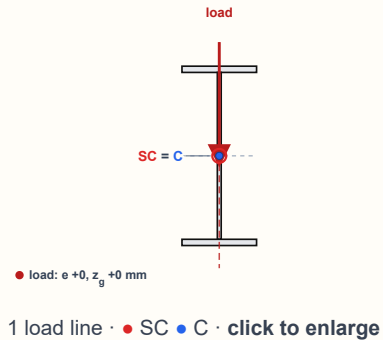
457 191 82 UB S275 ($p_y = 275 \text{ N/mm}$) $L = 8 \text{ m}$

Self-weight 0.8044 kN/m ? 0.8 m [Dead (G)] [automatic]

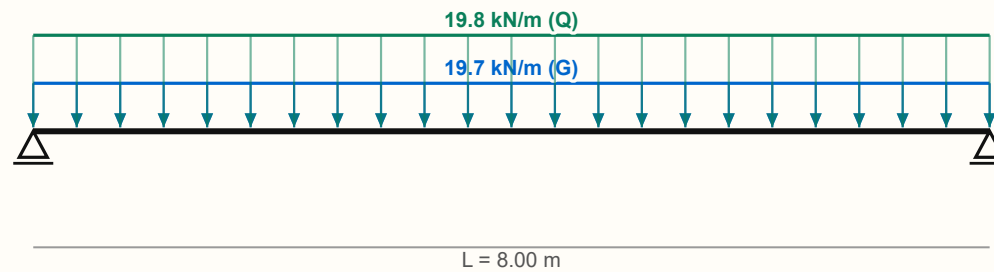
UDL 19.7 kN/m ? 0.8 m [Dead (G)]

UDL 19.8 kN/m ? 0.8 m [Imposed (Q)]

SECTION LOAD LINES



LOADING



Reactions (governing-moment combo, ULS: 1.35G + 1.5Q (Eq 6.10)): $R@0\text{m} = 229.52 \text{ kN}$ $R@8\text{m} = 229.52 \text{ kN}$

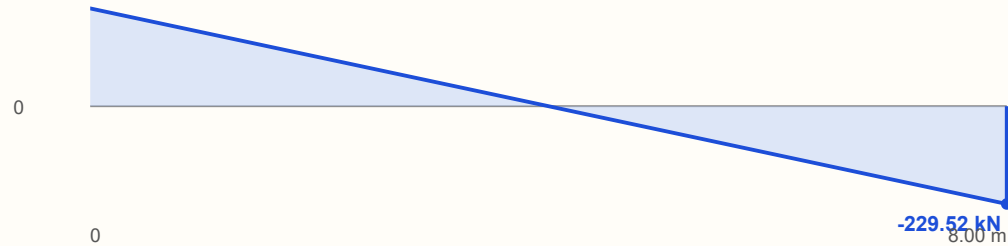
Load Combinations Considered

ULS combination	Max F_y (kN)	Max M_x (kN m @ m)	Governs
ULS: 1.35G + 1.5Q (Eq 6.10)	-229.52	459.05 @ 4	governs shear, governs moment

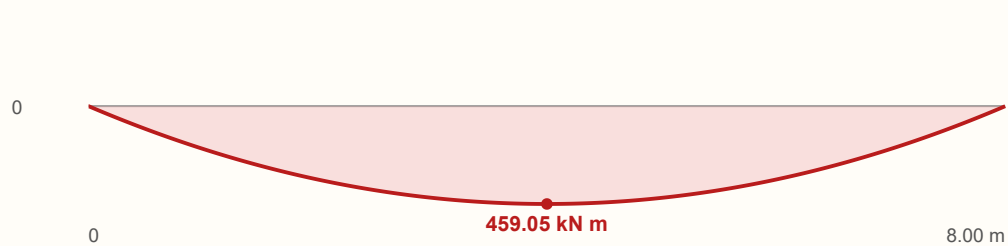
SLS combination	Deflection (mm)	Governs
SLS: Variable actions only (NA 2.23)	-13.55	governs deflection

Member Force Envelope ULS combinations Deflection SLS combinations			
Axial Force (kN)	Max Shear F _v (kN)	Max Moment M _x (kN m @ m)	Max Deflection (mm @ m)
0.000 C	-229.52	459.05 @ 4	-13.55 @ 4

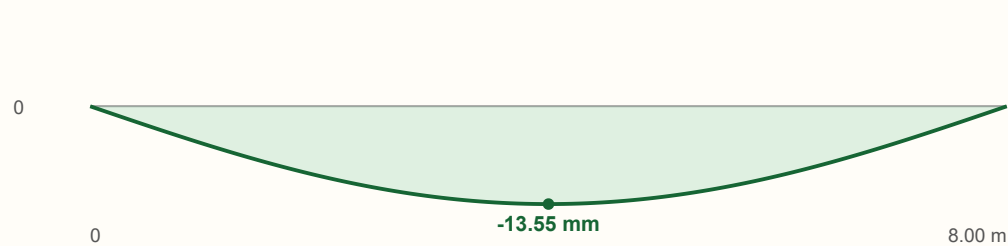
SHEAR FORCE (KN)



BENDING MOMENT (KN M)



DEFLECTION (MM, SLS)



Bending-moment diagram plotted on the tension (sagging-down) side. Shear and deflection plotted to true sign (down = below the axis).

Classification of Cross-Section (EN 1993-1-1 Table 5.2)

Section (82 kg/m)	457 191 82 UB [S275] h=460, b=191.3, t _w =9.9, t _f =16, r=10.2 mm		
Nominal yield strength f _y (Table 5.1 / BS EN 10025-2)	t _f = 16 mm (t ≤ 16)	275 N/mm²	f _u = 410
ε = √(235/f _y)	√(235/275)	0.92	
Outstand flange: c = (b – t _w – 2r)/2	(191.3 – 9.9 – 2×10.2)/2	80.5 mm	
c/t _f = 5.03	Class 1 limit: 9ε = 9×0.92 = 8.32	5.03 < 8.32	Class 1
Web: c = d = 407.6 mm; c/t _w = 41.2	Class 1 limit: 72ε = 72×0.92 = 66.56	41.2 < 66.56	Class 1
Section class under pure bending	max(flange, web)		Class 1

Shear Resistance (Cl. 6.2.6)

A _v = A – 2bt _f + (t _w +2r)t _f	10400 – 2×191.3×16 + (9.9+2×10.2)×16	4763 mm²	
but not less than ηh _w t _w (η=1.0)	1.0×428×9.9 = 4237 mm²	A _v = 4763 mm²	4763 > 4237
V _{c,Rd} = V _{pl,Rd} = A _v (f _y /√3)/γ _{M0}	4763×(275/√3)/1.0 ×10 ⁻³	756 kN	
V _{Ed} / V _{c,Rd} ≤ 1.0	229.5 / 756	0.3	OK

Shear Buckling (Cl. 6.2.6(6))

h _w /t _w ≤ 72ε/η	428/9.9 = 43.2 vs 72×0.92/1.0 = 66.6	43.2 < 66.6	Not required
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Moment Resistance (Cl. 6.2.5)

Shear at point of max. moment (Cl. 6.2.8)	V _{Ed} @ x=4 m: 0.0 kN vs 0.5V _{pl,Rd} = 378 kN (cl 6.2.8(2))	no reduction	Low shear
M _{c,Rd} = M _{pl,Rd} = W _{pl,y} ·f _y /γ _{M0}	1830×275/1.0 ×10 ⁻³	503 kN·m	
M _{y,Ed} / M _{c,Rd} ≤ 1.0	459 / 503	0.91	OK

Lateral–Torsional Buckling (Cl. 6.3.2.1)

Restraint condition	Beam fully laterally restrained — compression flange held in position throughout its length	LTB cannot occur	Not required
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Vertical Deflection of Beam (BS EN 1993-1-1 NA 2.23 — SLS: Variable actions only (NA 2.23))

w _{max} (governing SLS combination)	@ x = 4 m	13.6 mm	
Limit = span/360	8000/360 = 22.2 mm	13.6 < 22.2 mm	OK

ADVISORY (does not block PASS): Web bearing and buckling of the unstiffened web under concentrated loads and at supports (EN 1993-1-5 clause 6) are outside this calculator’s scope - verify separately wherever a point load or a reaction is applied to the web.

Fully restrained beam: design follows the SCI worked-example procedure to BS EN 1993-1-1 (UK NA) — classification (Table 5.2), shear resistance (cl 6.2.6), shear-buckling screen (cl 6.2.6(6)), moment resistance (cl 6.2.5, with the cl 6.2.8 shear check made at the point of maximum bending moment), and vertical deflection. γ_{M0} = 1.0 (UK NA); η = 1.0 taken conservatively.

Vertical deflection is checked against span/360 under the enabled SLS combination(s). NA 2.23 applies the limit to deflection due to **variable actions only**, so the default SLS combination carries Q only (edit under Load Combinations if a different criterion is required).

Analysis: 2-node Euler Bernoulli beam elements (direct stiffness); reactions exact, shear/moment by statics, deflection at nodes exact. Section data: SCI P363 Blue Book. This is a design aid results to be verified by a competent engineer.